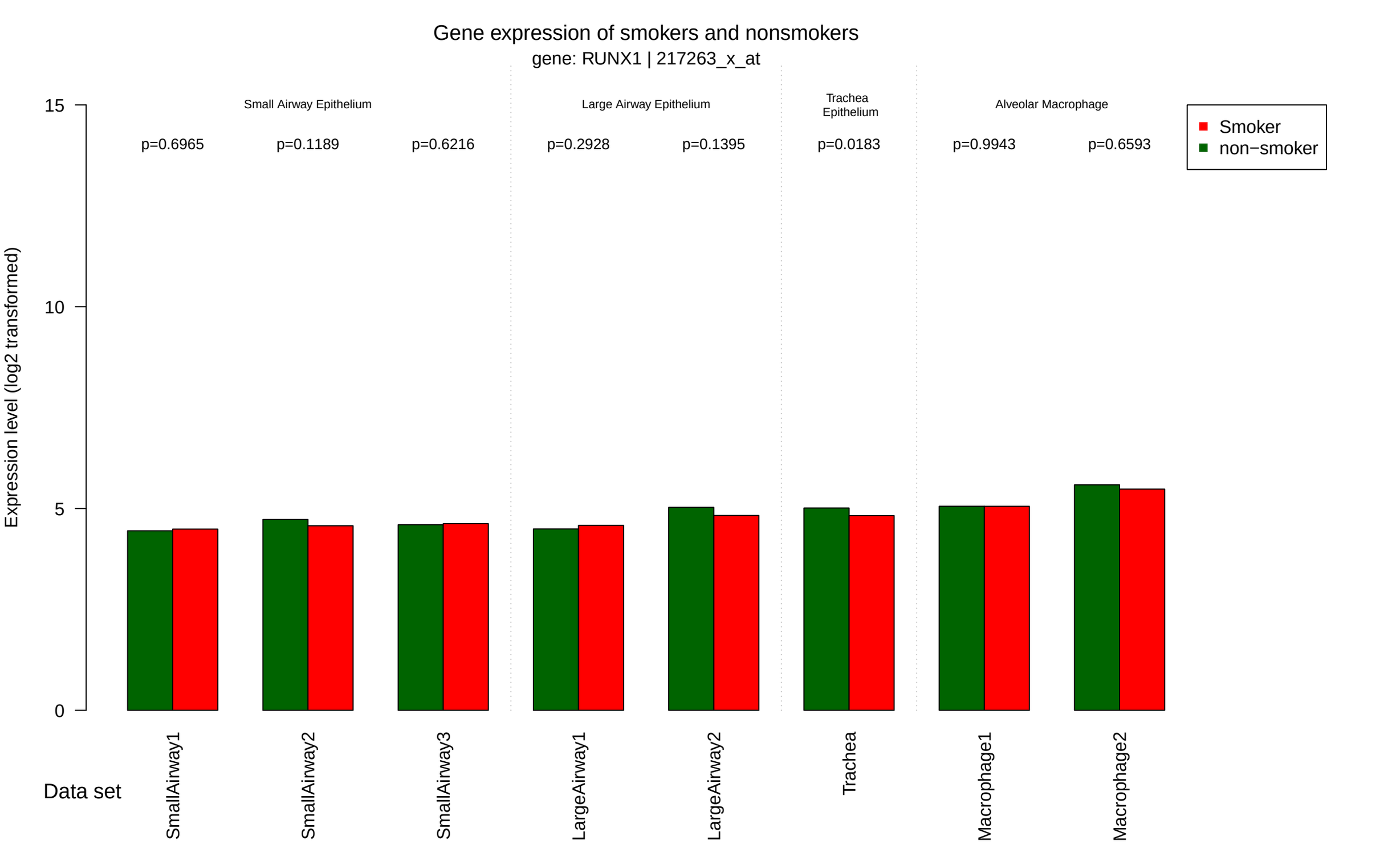
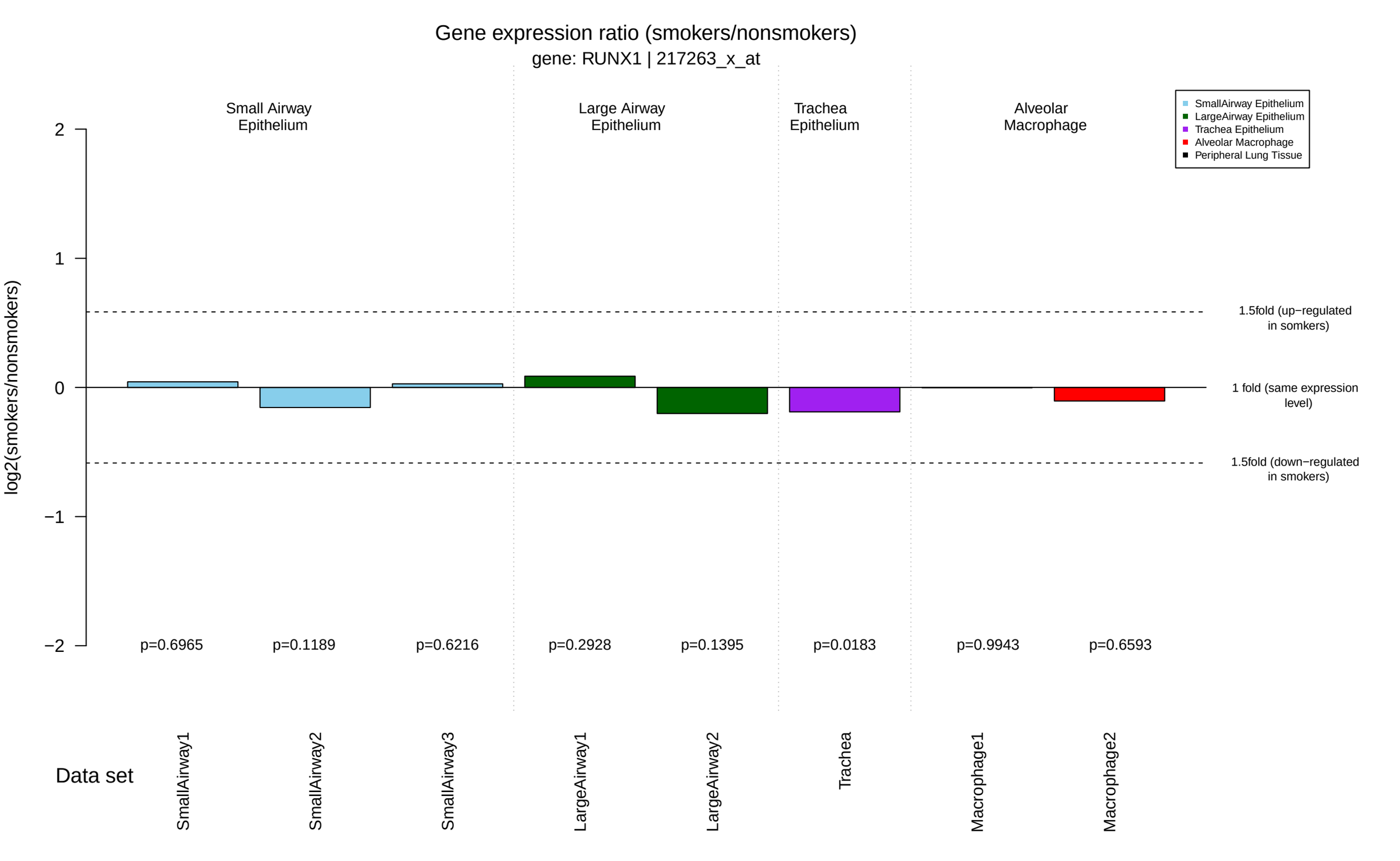
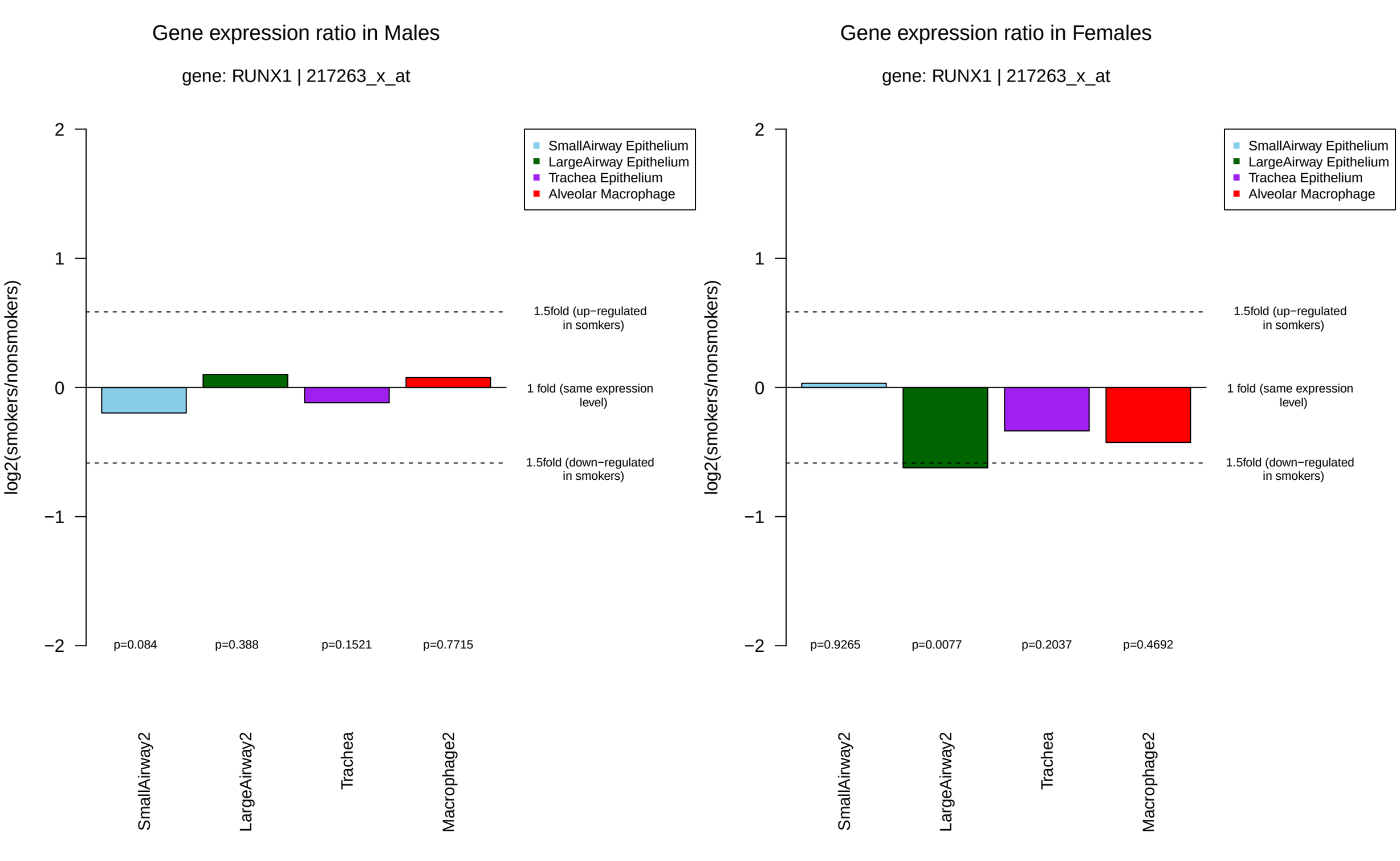
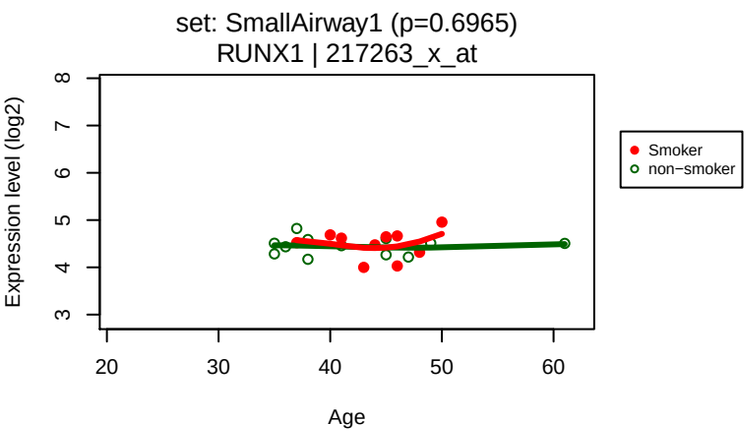
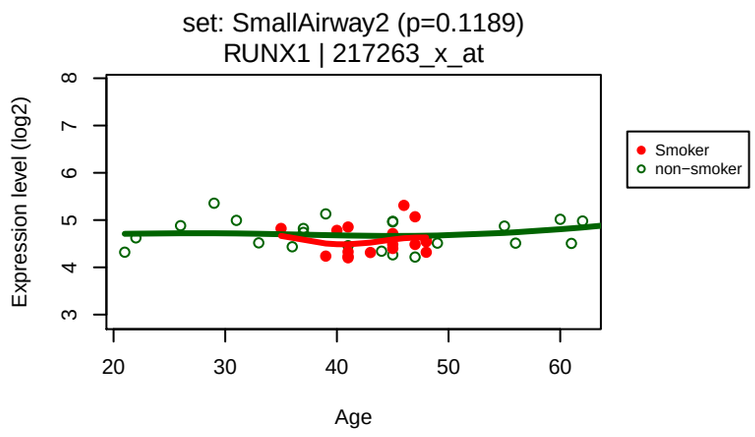
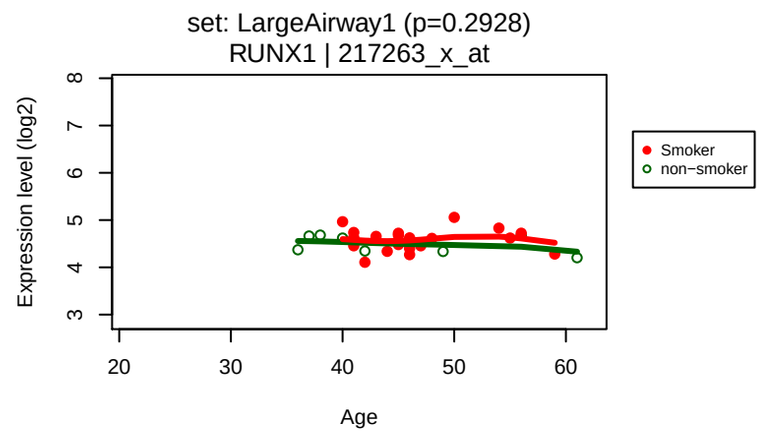

Gene expression of smokers and nonsmokers
gene: RUNX1 | 217263_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6965
p=0.1189
p=0.6216
p=0.2928
p=0.1395
p=0.0183
p=0.9943
p=0.6593
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

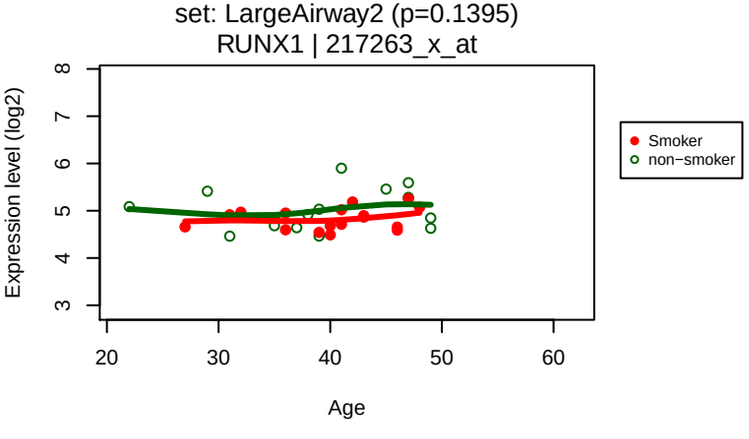

Gene expression ratio (smokers/nonsmokers)
gene: RUNX1 | 217263_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6965
p=0.1189
p=0.6216
p=0.2928
p=0.1395
p=0.0183
p=0.9943
p=0.6593
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

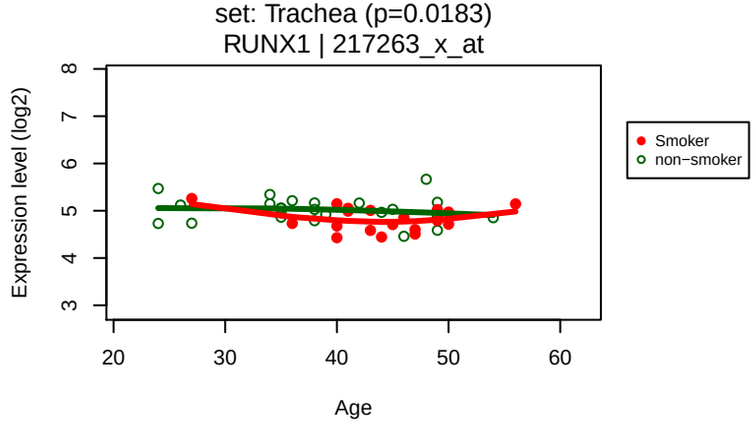

Gene expression ratio in Males
gene: RUNX1 | 217263_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.084
p=0.388
p=0.1521
p=0.7715
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RUNX1 | 217263_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9265
p=0.0077
p=0.2037
p=0.4692
SmallAirway2
LargeAirway2
Trachea
Macrophage2

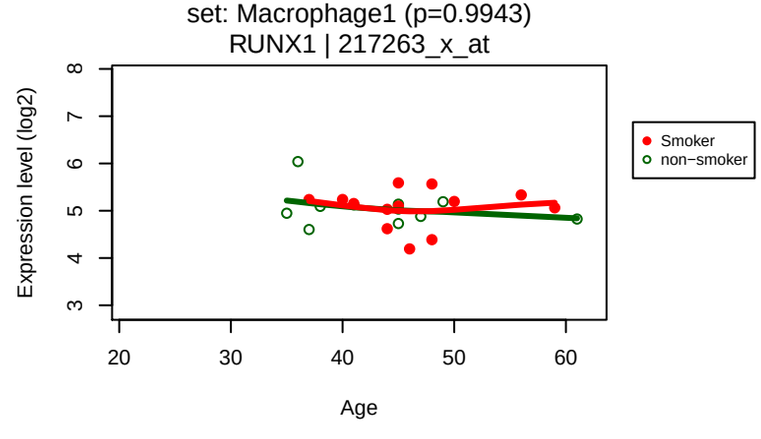

set: SmallAirway1 (p=0.6965)
RUNX1 | 217263_x_at
Expression level (log2)
Age
Smoker
non−smoker

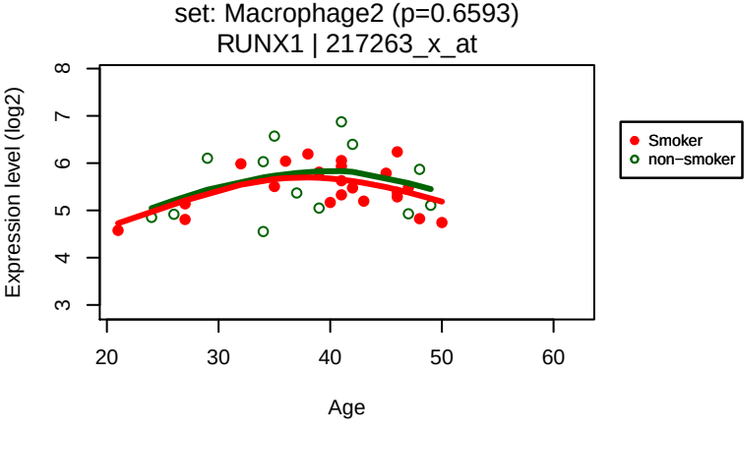

set: SmallAirway2 (p=0.1189)
RUNX1 | 217263_x_at
Expression level (log2)
Age
Smoker
non−smoker

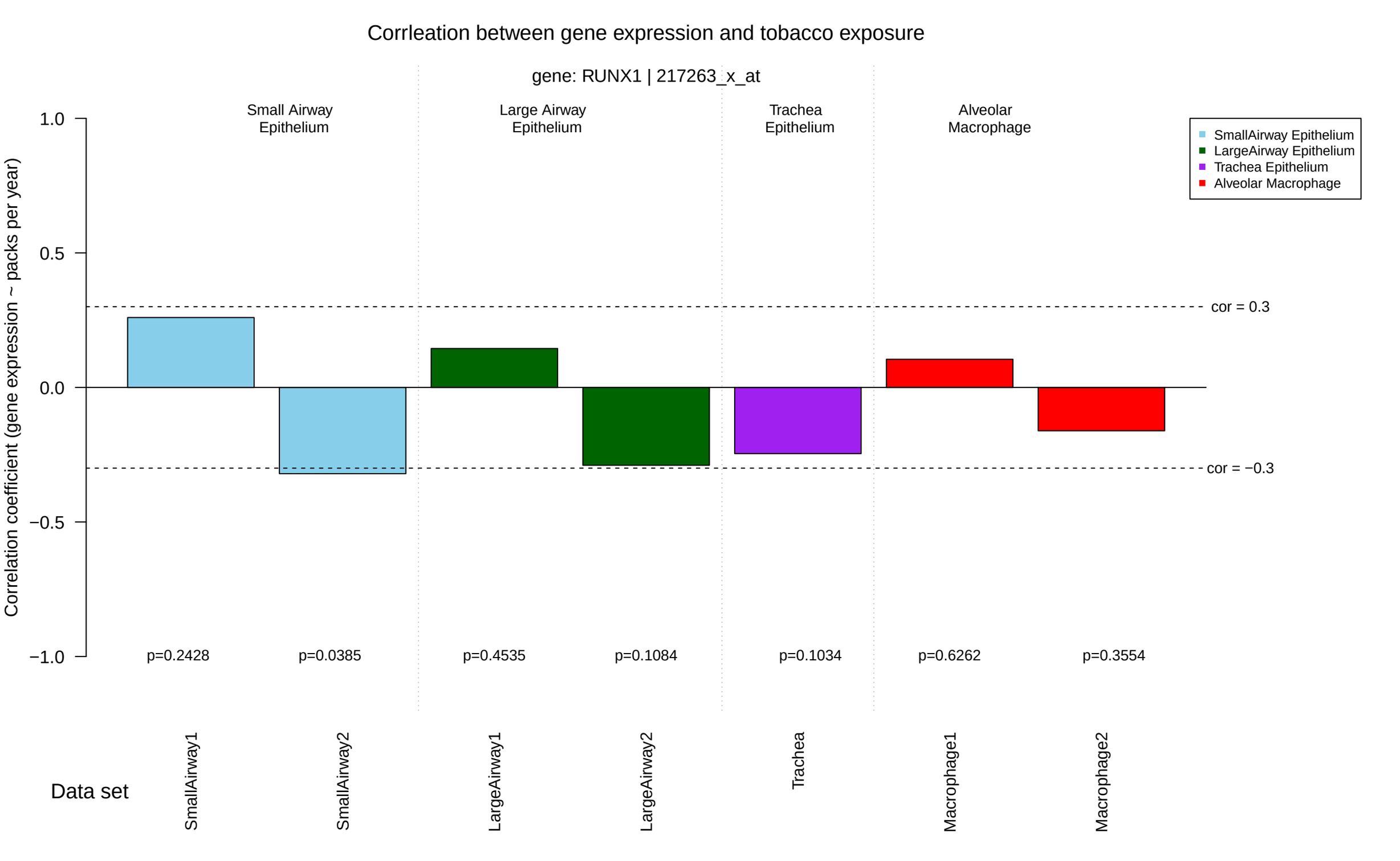

set: LargeAirway1 (p=0.2928)
RUNX1 | 217263_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1395)
RUNX1 | 217263_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0183)
RUNX1 | 217263_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9943)
RUNX1 | 217263_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6593)
RUNX1 | 217263_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RUNX1 | 217263_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2428
p=0.0385
p=0.4535
p=0.1084
p=0.1034
p=0.6262
p=0.3554
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2